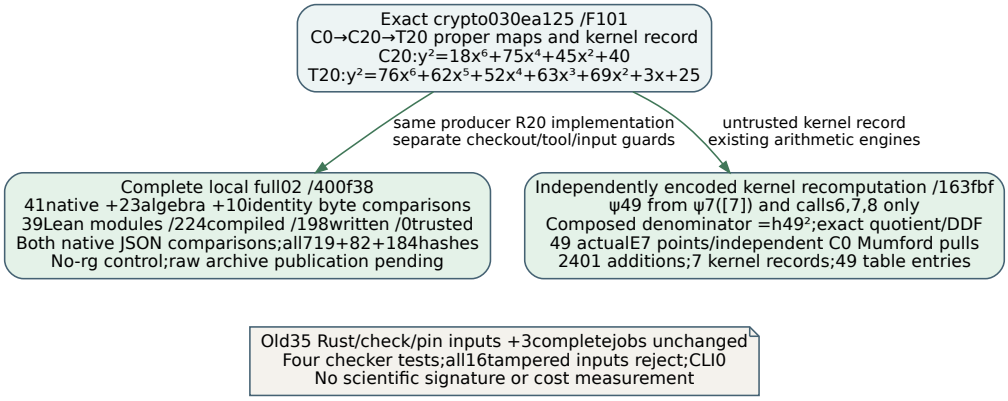
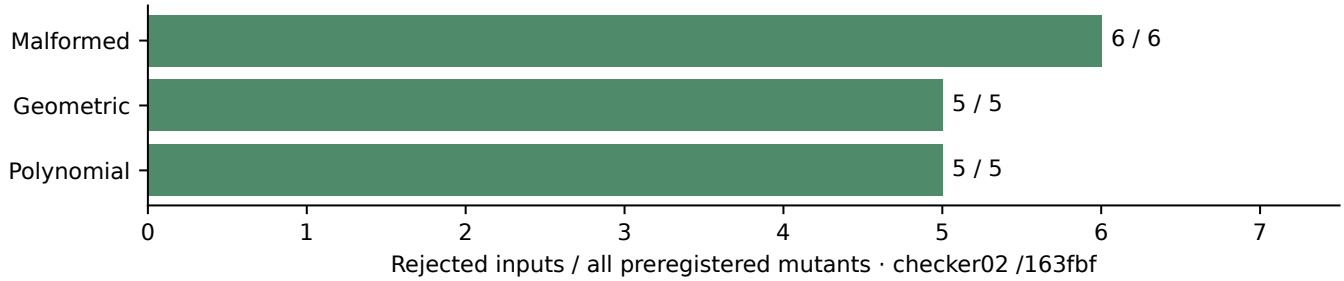
Proper T20 replay and independent kernel checks

2026-11-12 · source crypto030ea125 · consumer parent748e · unsigned

Complete local full02/400f38 passes41native/23algebra/ten identity byte comparisons,both native JSON comparisons and39proofmodules224compiled/ 198written0trusted. Four checker tests/all16mutants/CLI pass;all719upstream, 82consumer and184original hashes remain unchanged;rg is never called. Original start/finish/exit and guards complete. Whole158-file80516-byte archive is intact;remote raw publication is pending approved S3 migration. [pcc:unverified]



Scope:the proper-map replay executes the producer R20 implementation. The separate kernel verifier is independently encoded using the same existing arithmetic engines. Neither operation supplies a scientific signature. [pcc:unverified]



All deliberately constructed5polynomial/5geometric/6malformed mutants, not a probability sample. Three admission controls also reject2 before output:wrongHEADc7,modified newgeometryJSON and missinggrep (a5bf9e). The full positive replay uses an rg shim that always exits127 and must never be called. [pcc:unverified]

Derivation,faithful scope and retained attempts

For the fixed public E0=(1,46),E1=(26,58),E2=(59,27),rebuild the two normalized seven maps using the unchanged Kohel engine and exact cleared curve identities. Derive ψ49=ψ7·Hom24(ψ7;X7,ψ7²),using only division polynomial calls6,7,8. A divisor/leading-term argument proves the function composition formula;its Picard/function bridge is written in protocol75eacc, not fully formalized. Separately require the composed denominator=h49², monic degree24 h49,squarefreeness,exact ψ49 quotient,gcdψ7=h1 and Frobenius gcds giving3+21. No producer cyclic_kernel/construct_edges, Chart,pull or ψ49 recursion is called. [pcc:unverified]

Rebuild49 geometric E0[7] points and alpha images in the checked recorded degree12 field. Independent C0 pulls follow directly from X=x²+2,Y=y: in w=L/(x−a),use U=(a²−X+2)w²+2aLw+L²,V=Yw³/L modU. Derive the quintic by substitution and reuse the frozen Cantor engine by source path. Check49 distinct killed-by7 pulls,all2401 additions,seven actual native kernel Mumford records/all49 table entries and Frobenius5. Wrong-half42/7 is recomputed on alpha's elliptic images;it is distinct from the producer's target-Cantor check in the full suite. [pcc:unverified]

Upstream39 proof modules cover198 written/224 compiled declarations with zero trusted native. The actual curves,group order,Picard correspondence, polarization,Weierstrass interpretation and the division-function argument remain explicit written/native inputs. Existing ten IDC records use the unchanged checker at fixed2^61−1;that v1 format does not certify F101 coefficient equality in all characteristics. No new universal algebraic identity or independent R20 implementation is inferred from the kernel record recomputation. [pcc:unverified]

Checker01/f0c793 rejects the positive record because it treats sum-table entries as positions in7 kernel rows. The actual wire format stores indices in all49 E7 points. Fix02 requires global bound and actual kernel membership, then compares every actual Cantor sum;all49 checks remain. The failed source/lock/full log/ guards are immutable. Temporary harness syntaxfailure f956ff is caught before control execution and preserved before correction. The initial sparse clone warns about an unselected historical backlog blob; selected source closure remains fully guarded. [pcc:unverified]

Full01/4a80d5 exits124 while rebuilding the old engine in the checker five-minute step,after native41/ algebra23ten/bothJSONcmp/39proofmodules complete. Four checker tests/CLI/final driver guards are not reached;full01 is NOT accepted. Original183 afterhashes pass,late719upstream/81consumer checks are explicitly late. Entire159-file79644-byte standardtar.xz is intact,with member/raw/archive hashes and exact roundtrip;raw publication is pending because it exceeds64KiB. No file is pruned or split. [pcc:unverified]

Additive71036e reuses the exact engine dependency from this run's fresh isolated native Cargo target. Complete full02 retains all30/15/10/5minute bounds,counts/guards/byte comparisons/no-rg and reruns every phase. Root030e own43299283518 succeeds all4/94f85a;consumer748e's prior three-job ownCI is settled3ea63d. New consumer own four-job CI remains separate. [pcc:unverified]

LocalLinuxx86_64 development,Rust1.99.1/Lean4.35.0,no benchmark isolation or cost measurement. Trusted-base crypto artifact-autofix is the documented S3 route6b43e2;prior job is a no-op,clarified by root3eeb2e. Real new upload/replay/manifests are still required;metadata is not raw availability. Sources:Galbraith9.8.5–7,BD§4,root063773/720a59/b24be8,old replay9653c3. KB unavailable,actual127 receipts retained. REPORT.md links raw custody, code,protocol,editable sources and the inspected final PDF. [pcc:unverified]